AgriSense AI

Precision Agriculture Platform

AI-Powered Mobile Platform for Ethiopian Smallholder Farmers

5M+	95%	$200K	40%
Farmers Targeted	Disease Detection Accuracy	Funding Request	Yield Increase

AgriTech | Smart Farming | AI/ML | IoT

PROBLEM

Ethiopian smallholder farmers lose **30-40% of their crops** due to late pest detection, inefficient irrigation, and lack of real-time weather data. Traditional farming methods rely on manual inspection, causing delayed interventions and significantly reduced yields.

30-40%	5 Million	$2.5B
Crop Loss Rate	Farmers Affected	Agricultural Market Size

SOLUTION

An **AI-powered mobile platform** that combines computer vision, IoT sensors, and satellite data to deliver real-time agricultural intelligence directly to farmers via SMS and mobile app.

- **Crop Disease Detection** — Computer vision identifies crop diseases with 95% accuracy, enabling early intervention before spread.
- **Soil Moisture Monitoring** — IoT sensors track soil conditions in real-time, optimizing irrigation schedules and reducing water waste.
- **Weather Prediction** — Satellite data integration provides hyper-local weather forecasts for better planning.
- **Real-Time SMS Alerts** — Actionable recommendations delivered instantly, even on basic feature phones without internet.

TECHNOLOGY STACK

- **Computer Vision** — Deep learning models for disease/pest detection from crop images
- **IoT Sensors** — Low-cost soil moisture and environmental monitoring devices
- **Satellite Data** — Remote sensing for weather patterns and crop health monitoring
- **Mobile Application** — Cross-platform app with offline capability and multi-language support
- **SMS Gateway** — Universal alert system reaching farmers without smartphones

MARKET OPPORTUNITY

The platform targets **5 million smallholder farmers in Ethiopia**, with 1 million already having smartphone access. The solution is immediately expandable to East African agricultural markets including Kenya, Tanzania, and Uganda.

Market	Size	Access
Ethiopia Farmers	5 Million	1M Smartphones
East Africa Expansion	25+ Million	Growing Rapidly
Agricultural Market Value	$2.5 Billion	High Growth

BUSINESS MODEL

Freemium SaaS model designed for sustainable scaling across diverse farmer segments:

Tier	Price	Target	Features
Free	$0/month	All Farmers	Basic alerts, weather data, community access
Premium	$5/month	Individual Farmers	Disease detection, personalized recommendations, priority supp
Enterprise	$200/month	Cooperatives, Plantations	Multi-farm dashboard, bulk analytics, API access, dedicated supp
Data Insights	Custom	Agencies/Govt	Aggregated agricultural intelligence reports

INNOVATION HIGHLIGHTS

- **First AI platform** specifically designed for Ethiopian smallholder farmers
- **Offline capability** — Core features work without internet via SMS
- **Multi-language support** — Localized for regional languages and literacy levels
- **Low-cost IoT** — Affordable sensor hardware designed for developing market budgets
- **Scalable architecture** — Cloud-native platform ready for rapid regional expansion

SOCIAL IMPACT

40%	30%	100K+	50+
Yield Increase	Water Savings	Farmers Reached (Y2)	Jobs Created

TEAM

A multidisciplinary team combining agricultural expertise, AI/ML engineering, and local market knowledge:

- **Agricultural Experts** — Deep understanding of Ethiopian farming practices and challenges
- **AI/ML Engineers** — Computer vision and predictive modeling specialists
- **Local Partnerships** — Strategic alliances with agricultural cooperatives and government agencies

FUNDING REQUEST

$200,000 seed funding to develop MVP, conduct pilot programs, and achieve initial market traction.

TIMELINE

Phase	Duration	Milestone
MVP Development	3 Months	Core platform with disease detection and SMS alerts
Pilot Program	6 Months	Deployment with 100 farmers, feedback collection
Market Entry	12 Months	1,000 active farmers, premium conversions
Scale-Up	24 Months	10,000 farmers across Ethiopia, East Africa expansion planning

AI EVALUATION SCORECARD

Projected performance across key investment evaluation dimensions:

Dimension	Score	Rating	Assessment
Innovation	85/100	★★★★■	Novel AI + IoT combination for Ethiopian context
Market Potential	90/100	★★★★★	5M farmers, growing demand, regional scalability
Feasibility	78/100	★★★★■	Proven technology, adaptation required for local conditions
Social Impact	95/100	★★★★★	Direct food security improvement, job creation
Financial Plan	72/100	★★★■■	Freemium model needs significant scaling for profitability
Team Capability	75/100	★★★★■	Strong mix of technical and agricultural expertise
Overall Risk	25/100	★★■■■	Low risk — proven technology stack, clear use case
TRL Level	6/9	★★★★■	Technology validated in relevant environment

82-87 / 100

AgriSense AI — Empowering farmers with intelligent technology for a food-secure future.